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Petrochemical feedstock from pyrolysis of waste polyethylene and polypropylene using different catalysts

Bianca Pedroso S. Santos^a, Débora Almeida^a, Maria de Fatima V. Marques^{a,*},
Cristiane A. Henriques^b

^a Instituto de Macromoléculas Eloisa Mano, Universidade Federal do Rio de Janeiro – UFRJ, Av. Horácio Macedo, 2030, Centro de Tecnologia, Bloco J, Rio de Janeiro, Brazil

^b Instituto de Química, Universidade do Estado do Rio de Janeiro, Rua São Francisco Xavier, 524, Maracanã, Rio de Janeiro, Brazil

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ABSTRACT

In this work, thermal and catalytic pyrolysis of urban plastic waste composed of a mixture of polyethylene (PE) and polypropylene (PP) was studied. The catalysts investigated were HZSM-5, USY, NH₄ZSM-5 (ZSM-5 in ammonium form), and their corresponding alkaline-treated (BHZSM-5, BUSY, and BNH₄ZSM-5) or acid-leached (LHZSM-5, LUSY, and LNH₄ZSM-5) forms. The catalytic pyrolysis using BHZSM-5 (ZSM-5 in alkaline form), USY, BUSY, and NH₄ZSM-5 resulted in higher amounts of liquid fraction, which was composed of alkylbenzenes, naphthalenes, and olefins. Acid leaching of zeolite NH₄ZSM-5 increased both the specific area and the mesopore volume, allowing an increase in the liquid and gas fractions, and a reduction in the solid fraction. BHZSM-5 zeolite showed the best catalytic performance for the production of a high amount of liquid fraction, which was virtually composed of olefinic hydrocarbons without the formation of solid products. Moreover, BNH₄ZSM-5 zeolite (alkaline-treated NH₄ZSM-5) was a good pyrolysis catalyst also due to its high specific area and pore volume, resulting in higher gas formation in comparison with the liquid fraction (composed mostly of alkylbenzenes), and almost no solid fraction.

1. Introduction

Plastics make a fundamental contribution to society because of their versatility and relatively low cost. As a result, the amounts of synthetic plastic waste generated have been increasing steadily, causing environmental problems since these plastics do not degrade rapidly and can remain in the environment for long periods [1,2].

Another problem is that plastic waste is bulkier than organic waste, thus taking up more space in landfills, so adequate disposal is expensive. Indeed, the use of landfills is not an appropriate option to dispose of these wastes due to their low degradation rate [3–5].

Moreover, the incineration of these plastic wastes results in environmental problems due to increased emissions of harmful compounds such as sulfur and nitrogen oxides, dust, dioxins and other toxins, depending on the nature of the plastic [6,7].

Incineration and disposal in landfills cause the loss of chemical species that would be required to synthesize the polymers again [8]. Therefore, it is necessary to find other solutions that are more sustainable than incineration or disposal in landfills [9]. Numerous studies in the area of recycling and reuse of these polymers have been carried

out with the objective of producing raw materials and energy [10–12]. Various types of recycling have been found to be good options for controlling the increase of plastic waste because they are environmentally friendly compared to incineration and disposal in landfills. In fact, from recycling it is possible to recover raw materials, energy, and fuels in addition to minimizing the consumption of natural resources and inputs, and consequently reducing the environmental impacts of industrial activity [13–15].

Plastic recycling methods, according to ASTM D5033-00, are divided into four types depending on the final result obtained: primary (mechanical reprocessing into a product with equivalent properties), secondary (mechanical reprocessing into products requiring lower properties), tertiary (recovery of chemical constituents) and quaternary (recovery of energy). High-value polymers are normally mechanically recycled successfully, such as to make PET bottles. However, a major portion of plastics produced each year, composed essentially of polyolefins, is used to make disposable items for packaging or other short-lived products that are discarded within few months after manufacture. This indicates that the current use of plastics is not sustainable. Also, because of the durability of the polymers involved, substantial

* Corresponding author.

E-mail address: fmarques@ima.ufrj.br (M.d.F.V. Marques).

quantities of discarded plastics are accumulating in landfills worldwide.

On the other hand, in chemical recycling, chemical degradation leads to the production of liquid fuels and chemical compounds with a high added value from fragments of residual or segregated plastic material [1,16]. One of the methods of tertiary recycling is pyrolysis. This process can be thermal or catalytic and is a promising alternative that allows the conversion of low-value polymers into gaseous and liquid hydrocarbons [6].

Catalytic pyrolysis of plastics shows slightly higher levels of inert residue generation than mechanical recycling options, but much lower volumes than generated by other thermal and oxidative processes (incineration, gasification to methanol, gasification to syngas, gasification combined with combustion). Unlike the gasification processes, it produces a char, not a vitrified slag, which explains the higher hazardousness of its processing. Nevertheless, it is superior to the other thermal feedstock recovery processes in the cumulative energy demand balance (it is almost as energy efficient as conventional mechanical recycling processes) and ranks intermediate in the range of global warming potential, with lower CO₂ emission than achieved by other treatments [17].

The pyrolysis process is not expensive, and a wide range of final products can be obtained. In this process, heating occurs without the presence of oxygen. Depending on the material that will be pyrolyzed, organic compounds can be obtained as the final result, which will be cracked to generate mainly gaseous and liquid products. Also obtainable at the end of the process is an inorganic material which will remain unchanged in the solid fraction (char). The gas and liquid fractions can be used as a source of chemical products [18,19]. The catalyst has to be developed to reduce the char amount.

Thermal pyrolysis requires high temperatures, which usually results in products with low final quality, making this process unfruitful. This is because uncatalyzed thermal degradation gives rise to low molar mass substances, requiring further separation processes to result in new products [20,21].

This method can be improved by the addition of catalysts, which will reduce the temperature and the reaction time and allow the production of higher value-added hydrocarbons such as fuel oils and petrochemicals [22,23].

Both the physical and chemical properties of the catalysts will determine their cracking efficiency. These characteristics together promote the disruption of the C–C bonds and determine the type of product formed [24].

Catalyzed cracking has been widely used due to several advantages when it is compared to the thermal process since it more efficient concerning reactor use, reaction temperature, and residence time. However, catalytic cracking also has some disadvantages, such as the difficulty of recovering the catalyst after use, which increases its cost. Furthermore, the direct contact of the catalyst with the plastic residue deactivates it rapidly due to the deposition of matter and the poisoning by other elements such as chlorine, sulfur, and nitrogen contained in some additives that may be present in the plastic residue [25].

Thus, the development of more effective catalysts for the pyrolysis of polyolefins is essential to make pyrolysis more effective, allowing the formation of high-value chemicals [26,27].

The catalysts most used in the catalytic degradation of polymers are zeolites and amorphous materials [8]. The difference in the catalytic activity of these solids is mainly related to their acid sites, including their strength and quantity. The textural properties, such as specific surface area and pore volume, as well as particle size, also influence the performance of the catalytic process since these characteristics control the accessibility of large polyolefin molecules to the catalytically active internal sites of these solids [21,22].

In this work, two zeolites with different framework structure were selected for study: ZSM-5 and USY with MFI and FAU structure, respectively. ZSM-5 is a medium pore zeolite, which has a three-dimensional porous structure formed by a 10-MR straight channels

($5.6 \times 5.3 \text{ \AA}$) interconnect by 10-MR sinusoidal channels ($5.5 \times 5.1 \text{ \AA}$). It does not have cavities but the interconnections between the channels have a free diameter of about $9 \text{ \AA}$. This zeolite is thermally and chemically stable and possesses strong acid sites that are important for catalytic cracking. On the other hand, Y is a large pore zeolite whose three-dimensional framework structure is formed by sodalite cages connected by hexagonal prisms in a tetrahedral arrangement that forms supercavities ($\approx 11 \text{ \AA}$ diameter) which are interconnected by 12-MR apertures of about $7.1 \text{ \AA}$. USY zeolite (the ultra-stable form of Y zeolite) has mainly weak and medium strength acid sites, which are less effective in cracking than the strong ones. However, the presence of the supercavities with apertures of $7.1 \text{ \AA}$ favors the accessibility of the large plastic molecules to the acid sites located into the pore structure.

Thus, in the present work, the objective was to study the behavior of catalytic pyrolysis of urban plastic waste composed of polyethylene (PE) and polypropylene (PP) at 1:1 wt/wt ratio mixture using different types of zeolite as catalyst to obtain petrochemical inputs with high added value. The selected parent zeolites were ZSM-5, in both acid (HZSM-5) and ammonium (NH₄ZSM-5) forms, and USY. To the best of our knowledge, no study has been published regarding the comparison of micro/mesoporous ZSM-5 and USY zeolites for the catalytic pyrolysis of plastics, associating their catalytic properties with the PE:PP pyrolysis products distribution.

2. Experimental

2.1. Materials

Two types of municipal plastic waste (MPW) composed of post-consumed polyethylene (PE) and polypropylene (PP) were used in this study. They were obtained from a recycling company in the form of flakes and were washed three times with ethanol and dried overnight.

The parent zeolites investigated in the pyrolysis experiments were zeolite HZSM-5, provided by PQ Corporation in the acid form, ultra-stable Y zeolite (USY), and ZSM-5 in ammonium form (NH₄ZSM-5), supplied by Petrobras Research Center (CENPES). These zeolites were grounded and sieved and the fraction with particle size less than 200 mesh ($74 \mu\text{m}$) was separated for study, treated and used as catalysts for pyrolysis of plastics.

Samples of PE and PP were mixed with synthetic zeolites using a twin-screw mini-extruder in the counter-rotating mode at 60 rpm and 180°C for 7 min, obtaining samples weighing 5 g with well distributed catalyst particles in the polymer. The proportion was 1:1 wt./wt. in PE:PP mixture with and without zeolites. The amount of zeolite in the samples was 5% wt.

2.2. Treatment of the original zeolites

Mesoporosity can be introduced into the original zeolite samples through post-synthesis treatments such as acid leaching (L) and alkaline treatment (B).

For the alkaline treatment, the zeolite was suspended in a 0.2 mol/L NaOH solution using a zeolite (g)/NaOH solution (mL) ratio equal to 0.008, at 75°C , for 30 min, as previously established by Alves et al. [28]. After, the solid was separated by filtration and washed with deionized water until neutral pH. Then it was dried at 200°C , for 1 h. The alkaline-treated samples were converted into the acid (H) form by ion-exchange with a 2 mol/L NH₄Cl solution. Then, the zeolite was separated by filtration, washed with deionized water and calcined in a muffle furnace at 450°C , for 4.5 h. The samples were named Bzeolite, where B means the alkaline treatment.

The leaching treatment was carried out in two steps. In the first step, the zeolite was suspended in an 1 mol/L NH₄Cl solution using a zeolite (g)/solution (mL) ratio of 0.04, at 80°C , for 24 h. At the end of the treatment, the solid was separated by centrifugation, washed with distilled water and dried for 20 h at 110°C ; the solid was then heated in

a muffle furnace at 500 °C, for 4 h. In the second step, the obtained zeolite was refluxed with a 1 mol/L HCl solution for 5 h, and the solid was then separated by centrifugation, washed with distilled water until neutral pH and dried at 110 °C, for 20 h. The samples were named Lzeolite, where L means acid leaching treatment.

2.3. Pyrolysis of plastic residues

The plastic residue samples, mixtures of PE:PP (1:1 wt), with and without zeolites were submitted to the pyrolysis technique under nitrogen atmosphere at 450 °C for 30 min. The pyrolysis unit used consists basically of a vertical furnace and a stainless steel tubular reactor [29]. At the upper end of the reactor, the nitrogen gas flow (35 mL/min) is adapted to the inlet valve whereas a glass condenser immersed in an ice bath is coupled to the exit valve at the bottom of the reactor to collect the pyrolytic liquid. The system operates based on standard test procedure established in ASTM 3907–92 [30].

The liquid pyrolysis products were monitored by gas chromatography using a Varian 1200 L gas chromatograph equipped with a DB-5 capillary column (30 m × 0.25 mm × 0.25 µm) and FID detector.

Solid residues (the amount of catalyst was discounted) and pyrolytic liquid fractions were determined by weight and the gas fraction was calculated by weight difference (according to Eqs. 1–3).

$$L = WCA - WCB \quad (1)$$

$$S = WRA - WRB \quad (2)$$

$$G = TW - (L + S) \quad (3)$$

where L, S and G are liquid, solid and gas fractions, respectively; WCB and WCA are condenser weight before and after pyrolysis; and WRB, WRA and TW are reactor weight before pyrolysis, reactor weight after pyrolysis and total weight, respectively.

The reproducibility of the experiments was evaluated, and the experimental data are the mean values of three replications. Significance was tested with a 95% confidence interval.

2.4. Characterization of the zeolites

The crystalline phases of the samples was determined by X-ray powder diffraction (XRPD) in a Rigaku Miniflex diffractometer using Cu-Kα ($\lambda = 1.542 \text{ \AA}$) radiation at 30 kV, and 15 mA. The XRPD data were recorded in a 2θ range of 5–50° with an angular step width of 0.05° and a counting time of 1° (20)/min.

The nitrogen adsorption–desorption isotherms of the samples were measured at –196 °C in a Micromeritics ASAP 2020 equipment. The micropore volume was determined by the t-plot method and specific surface area by BET method. Mesopore volume was calculated from the adsorption branch of the isotherm using the Barret–Joyner–Hallenda (BJH) model.

The bulk chemical composition of the zeolites was determined by energy-dispersive X-ray spectroscopy (EDX) carried out in a Shimadzu model 720 spectrometer under vacuum for 320 s, all samples being in the powder form. The results were expressed in % wt of the elements. The molar amounts of Si and Al in the samples were obtained by dividing the % wt. of Si and Al by the respective molar mass and the bulk Si/Al molar ratio was calculated.

The framework Si/Al molar ratio and EFAL (extra-framework alumina) content of the studied zeolites were determined by magic angle spinning (MAS) nuclear magnetic resonance (NMR) of ^{29}Si and ^{27}Al , respectively. ^{29}Si MAS-NMR spectra were obtained with a Bruker Avance 400 spectrometer at resonance frequency of 79.2 MHz whereas the spectra of ^{27}Al MAS-NMR were acquired at 103.9 MHz in the same spectrometer.

The framework Si/Al molar ratio was calculated from ^{29}Si MAS-NMR spectra using the formula below, where n is the number of Al atoms in the second coordinating sphere of Si and I is the intensity of the peak

corresponding to the Si(nAl) environment (Eq. (4)).

$$\frac{\text{Si}}{\text{Al}} = \frac{4 \sum_{n=0}^{n=4} I_{\text{Si}(n\text{Al})}}{\sum_{n=0}^{n=4} n I_{\text{Si}(n\text{Al})}} \quad (4)$$

Two major peaks were present in the ^{27}Al MAS-NMR spectra. The first one, observed at a chemical shift of 60 ppm is related to the framework tetrahedral Al. The second one, observed at around 0 ppm corresponds to extra-framework Al (EFAL) with octahedral coordination (EFAL% = 100% – Al_{tetra}%).

3. Results and discussion

Initially, the original zeolites and the modified ones were characterized to determine their chemical composition (bulk and framework Si/Al molar ratio) and their textural properties. Then the materials were evaluated as catalysts for the pyrolysis of PE:PP mixture at 450 °C for 30 min.

3.1. Characterization of the zeolites

Table 1 shows the bulk and framework composition of the studied zeolites, expressed as Si/Al molar ratio, obtained via EDX and ^{29}Si MAS-NMR, respectively, as well as the amount of extra-framework alumina (EFAL) determined by ^{27}Al MAS-NMR. The standard deviation of the values is near 10%.

The bulk Si/Al molar ratio of the samples HZSM-5, NH₄ZSM-5 and USY are in accordance with those informed by the providers. The alkaline treatment (named B + zeolite) was carried out to desilicate the zeolite structure thus reducing its Si/Al ratio and generating mesopores. In contrast, the acid leaching treatment (named L + zeolite) remove both framework and extra-framework aluminum, thus raising the Si/Al ratio and also generating mesoporosity.

For HZSM-5 zeolite, the results show that the dealumination promoted by the acid leaching was not significant. Thus, only a small increase in both bulk and framework Si/Al molar ratios were observed. The stability of this sample to the acid leaching is typical of zeolites with a high framework Si/Al molar ratio. On the other hand, the silicon-rich framework of HZSM-5 led to a material more susceptible to desilication by alkaline treatment. In fact, a significant silicon removal was observed, reflecting in an important reduction in global and framework Si/Al values.

As to the sample NH₄ZSM-5, whose framework structure is aluminum-rich than that of HZSM-5, the acid leaching dealuminated the zeolite and an increase in both bulk and framework Si/Al molar ratio were observed for LNH₄ZSM-5. However, the presence of EFAL in a non-neglected amount suggests that part of the aluminum removed remained inside the porous structure of the zeolite. The alkaline

Table 1
Bulk and framework Si/Al molar ratio and EFAL content of the studied zeolites.

Catalyst	Si/Al bulk ^a	Si/Al framework ^b	EFAL ^c (%)
HZSM-5	41.0	38.0	2.92
LHZSM-5	42.5	42.4	1.30
BHZSM-5	6.92	9.43	n.d.
NH ₄ ZSM-5	12.9	17.5	n.d.
LNH ₄ ZSM-5	17.1	24.1	16.8
BNH ₄ ZSM-5	9.53	14.1	24.1
USY	2.83	4.53	51.7
LUSY	3.47	n.d.	71.7
BUSY	3.03	3.97	21.3

^a EDX.

^b ^{29}Si MAS-NMR.

^c ^{27}Al MAS-NMR.

treatment decreased both bulk and framework Si/Al molar ratios confirming the removal of framework silicon. The presence of an important amount of EFAL indicates that desilication was accompanied by the removal of structural aluminum. According to Song et al. [31], the removal of structural silicon atoms would reduce the stability of the neighboring Al–O bonds favoring aluminum removal.

The effect of alkaline treatment on the chemical composition of the sample BUSY was not very important and only a mild desilication of the framework was observed. Unexpectedly, the treatment with NaOH solution removed part of the EFAL present in the parent sample (USY).

The results observed for the alkaline-treated samples (BHZSM-5, BNH₄ZSM-5, and BUSY) confirm the supposition that there is an optimum Si/Al ratio of the parent zeolite to remove silicon from the framework by this method. For ZSM-5, it has been shown that the optimum is in the range 25–50 [32]. Higher amounts of framework aluminum close to silicon atoms difficult the removal of silicon since the negative charge of the AlO₄[−] tetrahedra hampers the hydrolysis of the Si–O–Al bonds in the presence of hydroxyl anions comparison to the relatively easy breaking of the Si–O–Si bonds without neighboring aluminum [33].

For sample LUSY, the peaks associated with Si(nAl) environment could not be detected and framework Si/Al molar ratio was not determined. This fact associated with the higher amount of EFAL suggests that acid leaching destroyed the framework structure of USY zeolite under the experimental condition studied.

Figs. 1–3 show the X-ray diffraction patterns of the studied samples along with the corresponding reference diffractogram of a pure and well-crystallized ZSM-5 (MFI structure) and Y (faujasite structure) zeolites [34]. Except for LUSY sample, the characteristic peaks of Y and ZSM-5 materials were observed in the X-ray diffractogram of the zeolites, which suggests that although the treatments carried out reduced the crystallinity of the samples (broader diffraction peaks), the framework structure was preserved, and no amorphous phase was formed.

On the other hand, no diffraction peaks were observed for LUSY sample, which confirmed the supposition inferred from the MAS-NMR analysis that the acid leaching destroyed the zeolite structure producing an amorphous material.

The influence of the acid and alkaline treatments on the textural properties of the studied zeolites is shown Table 2. It can be observed that, for ZSM-5 samples, the two procedures promoted an increase in both the BET specific area and the mesoporous volume, which is an important feature to facilitate the access of the bulk plastic molecules to the acid sites of the zeolites improving the catalytic activity for pyrolysis.

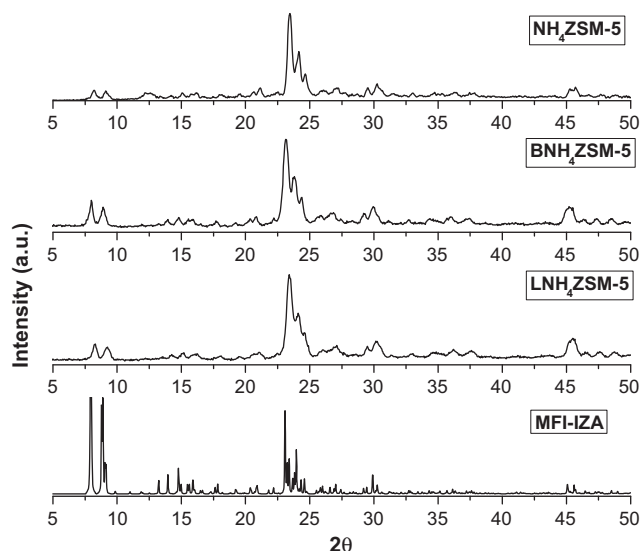


Fig. 2. Powder X-ray diffractograms of samples NH₄ZSM-5, LNH₄ZSM-5, BNH₄ZSM-5, and reference diffractogram for ZSM-5 structure from IZA database [34].

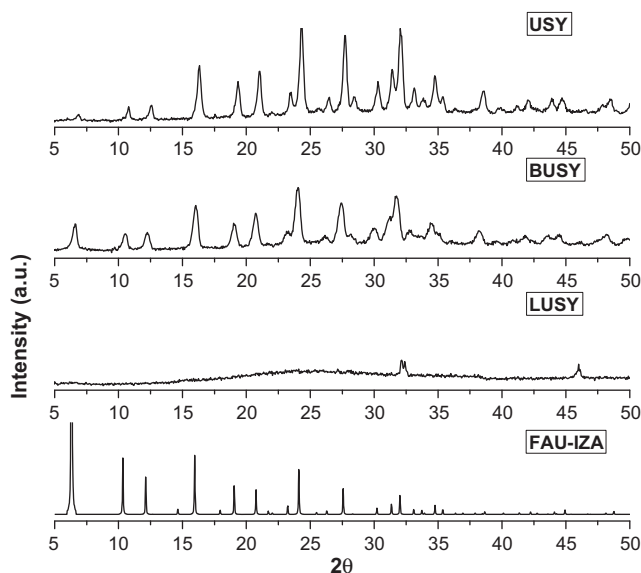


Fig. 3. Powder X-ray diffractograms of samples USY, LUSY, BUSY, reference diffractogram for faujasite structure from IZA database [34].

Table 2
Textural properties of the studied zeolites.

Sample	BET specific area (cm ² g ^{−1}) ^a	V _{meso} (cm ³ g ^{−1}) ^b	V _{micro} (cm ³ g ^{−1}) ^c
HZSM-5	369	0.11	0.14
BHZSM-5	410	1.16	0.09
LHZSM-5	692	0.70	n.d.
NH ₄ ZSM-5	325	0.05	0.16
BNH ₄ ZSM-5	372	0.24	0.11
LNH ₄ ZSM-5	793	0.46	0.09
USY	503	0.05	0.22
BUSY	580	0.10	0.24
LUSY	86	0.19	0.01

^a BET method.

^b BJH method.

^c t-Plot method.

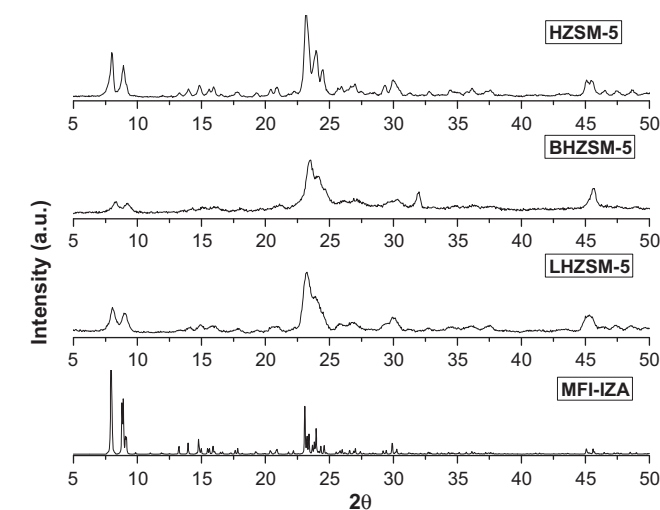


Fig. 1. Powder X-ray diffractograms of samples HZSM-5, LHZSM-5, BHZSM-5, and reference diffractogram for ZSM-5 structure from IZA database [34].

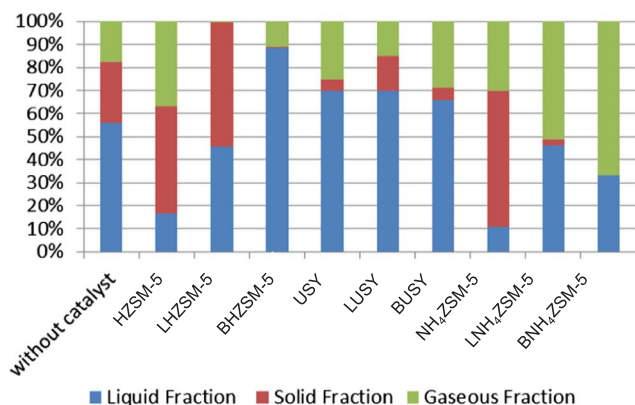


Fig. 4. Distributions of products by pyrolysis PE:PP at 450 °C for 30 min.

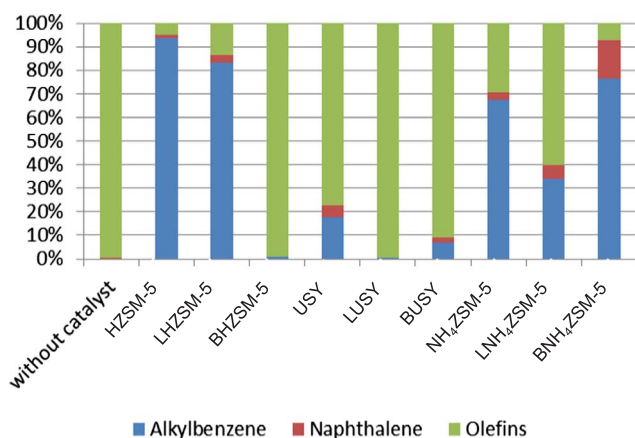


Fig. 5. Distribution of the liquid products of the pyrolysis of PE:PP at 450 °C for 30 min.

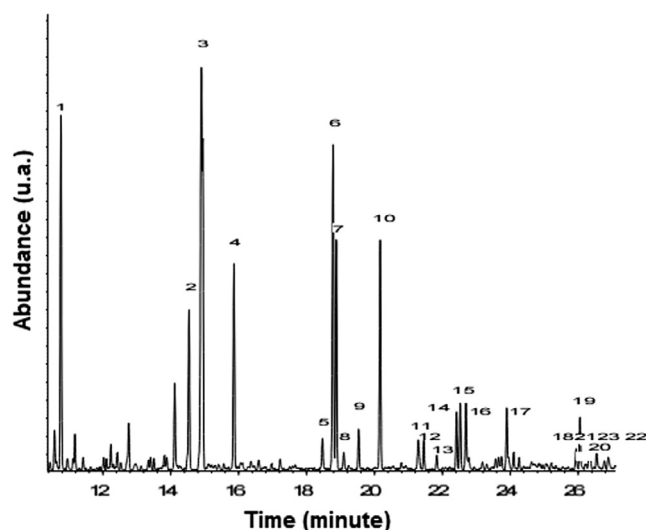


Fig. 6. Total ion chromatogram of the liquid fraction produced by the pyrolysis of PE:PP catalyzed by BNH₄ZSM-5 at 450 °C for 30 min. (1) toluene (2) ethyl benzene (3) m + p xylene (4) o-xylene (5, 6, 7, 8, 9, 10) C₉ alkyl benzenes (11, 12, 14, 15, 16, 17) C₁₁ alkyl benzene C₁₀ (13) indane (18, 20)-C₁₁ alkyl benzene C₁₁ (19) Methyl indane (21, 22) C₁₁ + alkyl benzene methyl indane (23) C₁₁ alkyl benzene tetralin.

For LUSY sample, the collapse of the framework structure due to the acid leaching produced an amorphous material with the lowest BET specific area and without microporosity. On the other hand, for BUSY sample, the slight increase in the textural properties is probably related to the removal of the EFAL species present on the precursor.

3.2. Pyrolysis experiments

The catalytic pyrolysis of the plastic mixtures containing zeolites are presented in Fig. 4 evidencing the effect of the addition of 5% wt. catalyst obtained after zeolite treatment on the yields of gas, pyrolytic liquid and residues (char).

Pyrolysis at 450 °C for 30 min favored the disruption of the C–C moieties of the polyolefins, generating hydrocarbons with smaller carbon chains, which constitute the liquid and gas products. Thermal pyrolysis resulted in high percentage of the liquid fraction (Fig. 4). In comparison to thermal pyrolysis, BHZSM-5 zeolite showed the best catalytic performance to the production of liquid products, without the formation of solid fraction (char). Besides, USY, BUSY and LN₄ZSM-5 also produced high amounts of liquid fraction and low solid content.

The comparison of the HZSM-5 family (parent HZSM-5, LHZSM-5, and BHZSM-5) shows that BHZSM-5 produced the highest liquid fraction (88%) among these catalysts and even the thermal process. Although both LHZSM-5 and BHZSM-5 present higher specific area and mesoporous volume than HZSM-5, the results show that the formation of solid residues (char) was favored over the acid leached zeolite (LHZSM-5). Thus it can be inferred that although the increase in specific area and mesoporous volume played an important role in the pyrolytic cracking, the framework composition must also be taken into account. Our results allowed us to suggest that the formation of char (solid residues) is favored over the zeolite with higher framework Si/Al molar ratio, which has the strongest acid sites. On the other hand, a higher density of acid sites with a lower strength (low framework Si/Al molar ratio) seems to improve the production of lighter fractions (liquid + gas).

As to NH₄ZSM-5 family, the production of liquid and mainly gaseous fractions was favored over LN₄ZSM-5 and BNH₄ZSM-5 zeolites, the latter producing the highest amount of gaseous products among the studied catalysts. Differently to that observed for LHZSM-5, the acid-leached LN₄ZSM-5 produced a higher amount of gaseous fraction and virtually no char. These results reinforce the supposition that besides the increase in specific area and mesoporous volume, the presence of higher amount of low strength acid sites (related to lower framework Si/Al molar ratio) favors the production of lighter fractions (liquid + gas). The results observed for USY and BUSY zeolite are also in accordance with this supposition.

When the parent zeolites (HZSM-5, NH₄ZSM-5, and USY) were compared, it can be observed that USY produced a higher amount of liquid fraction and a lower amount of solids than HZSM-5 and NH₄ZSM-5. Marcilla, Beltrán, and Navarro [35] evaluated the catalytic pyrolysis over HZSM-5 and USY zeolites and reported similar trends. According to the authors, these results could be related to the lower intrinsic acidity and the higher specific area and pore size of USY zeolite when compared to those of ZSM-5.

Based on our results, it can be concluded that both textural properties (specific area and mesoporous volume) and framework composition (Si/Al ratio) played an important role in the performance of the zeolite as a catalyst for the pyrolysis of PP:PE mixture. The higher production of lighter fractions (gas + liquid) was observed for the alkaline-treated ZSM-5 zeolites (BHZSM-5 and BNH₄ZSM-5). This type of treatment promoted the removal of silicon from the zeolite structure increasing the density of low strength acid sites (associated with the decrease in the framework Si/Al molar ratio) and generating mesopores. These two features seem to favor the formation of lighter products. Thus, BHZSM-5 zeolite was the most efficient catalysts for the production of liquid products whereas BNH₄ZSM-5 is the most promising choice for the production of gaseous ones.

The composition of the liquid fractions of the pyrolysis of PP:PE with and without catalyst was determined by gas chromatography, and the results are shown in Fig. 5. It can be observed that these fractions were formed by alkylbenzenes, naphthalenes, and olefins. The percentage of these groups was determined through the sum of relative peak

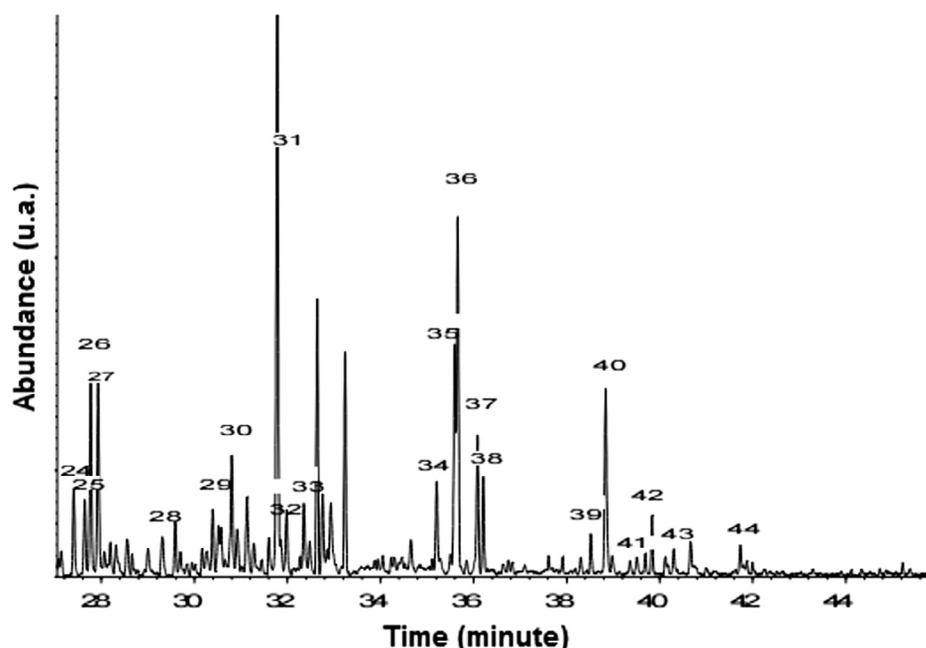


Fig. 7. Total ion chromatogram of the liquid fraction produced by the pyrolysis of PE:PP catalyzed by $\text{BNH}_4\text{ZSM-5}$ at 450°C for 30 min. (24) alkyl benzene C_{11} (25) naphthalene (26, 27) dimethyl indane (28) alkyl benzene C_{12} (29) dimethyl indene (30) methyl tetraline (31, 33) methyl naphthalene (32) Trimethyl indane (34, 35, 36, 37, 38)- C_2 naphthalene (39, 40, 41, 42, 43)- C_3 naphthalene (44)- C_4 naphthalene.

areas in each class of products identified with the use of analytical standards for gas chromatography. Figs. 6 and 7 present the total ion chromatogram for the liquid fraction produced over $\text{BNH}_4\text{ZSM-5}$ zeolite with the identification of the different products in each class.

Fig. 5 shows that the thermal pyrolysis produced only olefins in the liquid fraction. On the other hand, the composition of these fractions produced from the catalytic pyrolysis depended on the zeolite used as catalyst.

When the zeolites of the same family were compared (HZSM-5, LHZSM-5, and BHZSM-5; $\text{NH}_4\text{HZSM-5}$, $\text{LNH}_4\text{ZSM-5}$, $\text{BNH}_4\text{ZSM-5}$; or USY, LUSY, and BUSY) a fairly good correlation between the amount of olefins in the liquid fraction and the mesoporous volume (Table 2) could be found. Thus, for HZSM-5 family, BHZSM-5 with the highest mesoporous volume produced the highest amount of olefins whereas for HZSM-5 alkylbenzenes predominate in the liquid fraction.

According to Manos et al. [36] and Zhang et al. [37], the smaller mesoporous volume restricts the access of the bulky molecules produced as intermediate at the first step of pyrolytic cracking to the acid sites located inside the porous structure of the zeolite. This fact impairs subsequent cracking steps. Consequently, these molecules react preferentially at the surface acid sites, favoring the bimolecular reactions that produce aromatic compounds. Similar trends were reported by Seo et al. [38] that explain the results taking into account the textural and structural properties of the zeolites.

For BHZSM-5 sample, which has the highest mesoporous volume, the intermediate bulky molecules could access the acid sites inside the pore structure thus favoring their subsequent cracking that increases the olefin production.

The correlation between mesoporous volume and olefin formation was also observed for $\text{NH}_4\text{ZSM-5}$ family ($\text{NH}_4\text{HZSM-5}$, $\text{LNH}_4\text{ZSM-5}$, $\text{BNH}_4\text{ZSM-5}$). $\text{NH}_4\text{ZSM-5}$ has the lowest mesoporous volume, thus generating a higher amount of alkylbenzenes. Due to its higher pore volume, $\text{LNH}_4\text{ZSM-5}$ gave rise to a more significant fraction of olefins.

Finally, the effect of the higher mesoporous volume was also detected for USY and BUSY samples. BUSY zeolite produced more olefins than the parent USY due to its higher mesoporous volume. On the other hand, only olefins have been formed in the liquid fraction over LUSY sample. Since this material is an amorphous solid without a defined pore structure, its low acidity in comparison with the zeolites could explain the product distribution observed.

4. Conclusions

Both acid leaching and alkaline treatment were effective for the production of mesopores in ZSM-5 zeolites. Both textural properties (specific area and mesoporous volume) and framework composition (Si/Al ratio) played an important role in the performance of the zeolite as a catalyst for the pyrolysis of PP:PE mixture at 450°C for 30 min. The higher production of lighter fractions (gas + liquid) was observed for the alkaline-treated ZSM-5 zeolites (BHZSM-5 and $\text{BNH}_4\text{ZSM-5}$). This type of treatment promoted the removal of silicon from the zeolite structure increasing the density of low strength acid sites and generating mesopores. These two features seem to favor the formation of lighter products. As to the production of liquid hydrocarbons, catalysts that showed good results were BHZSM-5, USY, BUSY and $\text{LNH}_4\text{ZSM-5}$. The liquid fraction produced was composed of olefins, alkylbenzenes and naphthalenes with the predominance of olefins for the majority of the catalysts studied. A good correlation between the mesoporous volume and the amount of olefins in the liquid fraction was observed for the three families of zeolites studied. It was associated to the fact that for higher mesoporous volume the intermediate bulky molecules could access the acid sites inside the pore structure thus favoring their subsequent cracking that increases the olefin production. To obtain a higher amount of liquid fraction a zeolite with a higher mesoporous volume is preferable, whereas for the production of gaseous fraction a zeolite with an intermediate volume of mesoporous give better results. Thus, using an appropriate catalyst and post-synthesis treatment it is possible to obtain specific products with high added value.

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